

Internship Design Proposal

Intern: Aqsa Aslam

Program: CoderAxo Web Development Internship

Offer ID: CAX-OL-2026-310

Submission Date: June 22, 2026

Career Goal

In terms of where I see myself heading within Computer Science, my focus is Software Engineering, and specifically a Java Developer role further down the line. Alongside that, I'm building up practical full-stack development experience — currently through the MERN stack — since the backend, database, and architecture thinking involved carries over well regardless of language. I also take on freelance web projects on the side, which keeps these skills active outside of a classroom setting.

Learning Objectives

What I'm hoping to get out of this internship is real, applied experience rather than just theory — writing front-end code that's actually maintainable, structuring backend services properly with Node.js and Express.js, designing a MongoDB database that holds up under real use, and putting together React interfaces that are genuinely component-driven rather than one giant file. On top of that, I want to get better at API design, basic security practices around authentication, and documenting a project the way a real team would expect.

Organization Overview & Role

CoderAxo runs a structured, cohort-based Web Development Internship built entirely around building things rather than sitting through lectures alone. The program moves through the stack in stages — HTML/CSS/JS first, then Node.js, then React — with a mentor running live sessions and reviewing what's submitted each week. My part in this, as a Web Development Intern, is to work through each stage's assigned tasks, keep up with the weekly reporting requirements, and in the final stretch, contribute to building a complete, deployable MERN application under a fixed deadline.

Key Activities

- Coding
- Testing
- Documentation

Tools / Technologies to be Used

HTML5, CSS3, and modern JavaScript (ES6+) on the front end; React.js for building interfaces; Node.js and Express.js on the backend; MongoDB for data storage; Tailwind CSS for styling; Git and GitHub for version control; JWT and bcrypt for authentication and password security; Vite as the build tool; and Postman for testing APIs along the way.

Expected Deliverables

Across the internship, the expected outputs are weekly activity logs and progress reports, three individually assessed tasks (a personal portfolio site, a JavaScript-driven To-Do app, and a deployed React application), and, as the capstone, a fully built MERN-stack project — the Project & Team Collaboration System — handed in with its source code, an ER diagram/schema, a README, API documentation, a short demo video, and a written final report.

Academic Preparation

This work builds directly on courses from my BS Software Engineering degree at COMSATS University Islamabad — Software Construction & Development, Object-Oriented Programming, Database Systems, and Web Engineering in particular — as well as prior project and internship exposure to web and software development.

Additional Comments

I see this internship less as a detour from my Java Developer goal and more as groundwork for it — the same backend logic, database design, and system-level thinking apply regardless of which language sits on top. Glad to have a structured, deadline-driven setting to actually put that into practice.